

Lanternfield Goods — Okta IAM lab report

Previous: [Leaver leaves Rostr active](#).

Date: 2026-10-07. The work so far took three days, 5 October through 7 October. Clocked time is in [docs/worklog/week-1.md](#). This is a learning lab, not production Okta experience. Every claim below points at a file in this repository. The PDF at the repo root is an export of this page. The three practice accounts are [docs/interview-stories.md](#). The CV lines are not written up here, and the org has not been torn down.

Start here

Lanternfield Goods is a fictional shop. Its staff list is a JSON file in Git. Okta is the directory: it holds the user accounts, the groups, and the sign-in rules. Rostr is a small rostering app built for this lab, so both sides of every sign-in and every account update can be read. Nothing here is a real employer or a real customer.

A joiner is a new hire. A mover is a person who changes department or job. A leaver is a person who leaves. The HR file is the only place those changes are typed in. A script reads the file and calls an Okta workflow. Okta then creates or updates the account inside Rostr.

Word	Meaning in this lab
SAML	Staff sign-in. Okta posts a signed message to Rostr. The address that receives it is the ACS URL, <code>/saml/acs</code> .
OIDC	Admin sign-in. Okta returns a token. Rostr reads the groups in that token before it opens <code>/admin/users</code> .
SCIM	The API Okta uses to create, update, and deactivate the Rostr account. It is not the sign-in.
Group rule	Membership filled from a profile field. Change the department, and the department group follows. A membership added by hand does not.
APP-Rostr-Users	The group that is allowed to open Rostr.
APP-Rostr-Admins	The group that is allowed to open the admin page. It is empty unless someone was granted it.

Person	Why they appear
Jonah Hale	The staff user used for sign-in tests. He started in Sales and was later moved to Operations.
Priya Shah	The clean department move, Sales to Operations.
Samir Adeyemi	The first leaver. Deactivated in Okta and in Rostr.
Thomas Okeke	The joiner. Later deactivated in Okta while Rostr was left active on purpose, then deactivated in Rostr on a second try.

Lena Ortiz	Filed the one-hour request for admin access. Ticket 357784 was closed by Admin on 8 October.
Drill Scim	A throwaway user for the provisioning failure. Deactivated afterwards. Not in the HR file.

Purpose

The point is to practise the operational work of an identity engineer: day-to-day Okta administration, connecting an app with SAML and OIDC, provisioning accounts with SCIM, automating joiner, mover, and leaver, and keeping the evidence an auditor would ask for.

The design is [docs/decisions/o2-design.md](#). The session record is [docs/worklog/week-1.md](#).

How it was built

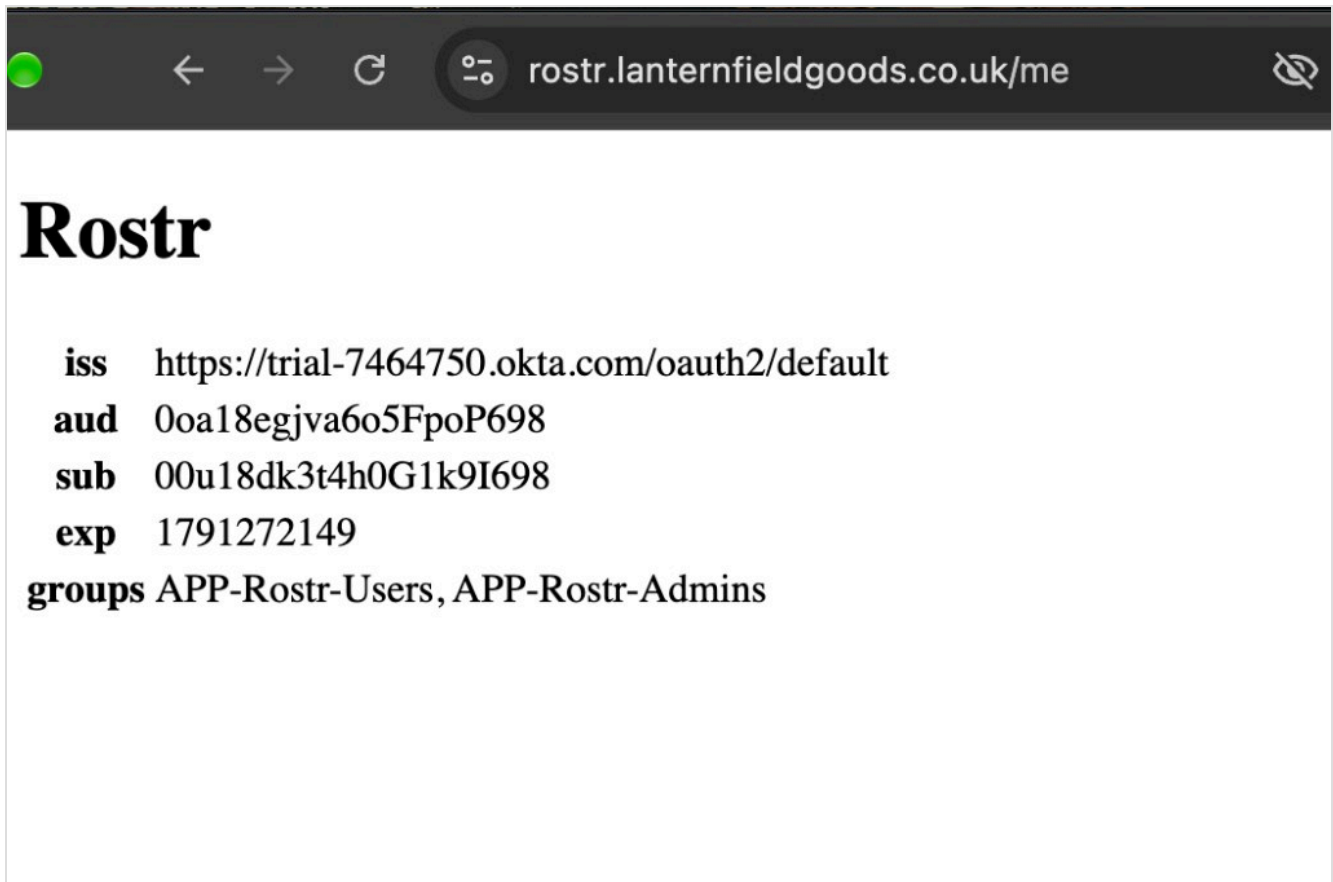
The domain is `lanternfieldgoods.co.uk`, not the `.com` name in the runbook. Cloudflare Email Routing receives mail and does not send it. Rostr's public name is `rostr.lanternfieldgoods.co.uk`, a named Cloudflare Tunnel to the container on port 3000. The tunnel credentials are outside the repo. [docs/decisions/00-domain.md](#).

The org is `trial-7464750`, a 30-day Workforce Identity free trial. It is not the Integrator Free Plan the design assumed. Lifecycle Management on this trial ends when the 30 days end. The 10-active-user cap and the five-Workflow cap were kept anyway. [docs/decisions/01-org.md](#).

Two administrator accounts exist. The daily admin does the work. Break-glass is a separate Super Administrator with its own authenticator, created before the staff policies. [docs/decisions/02-break-glass.md](#).

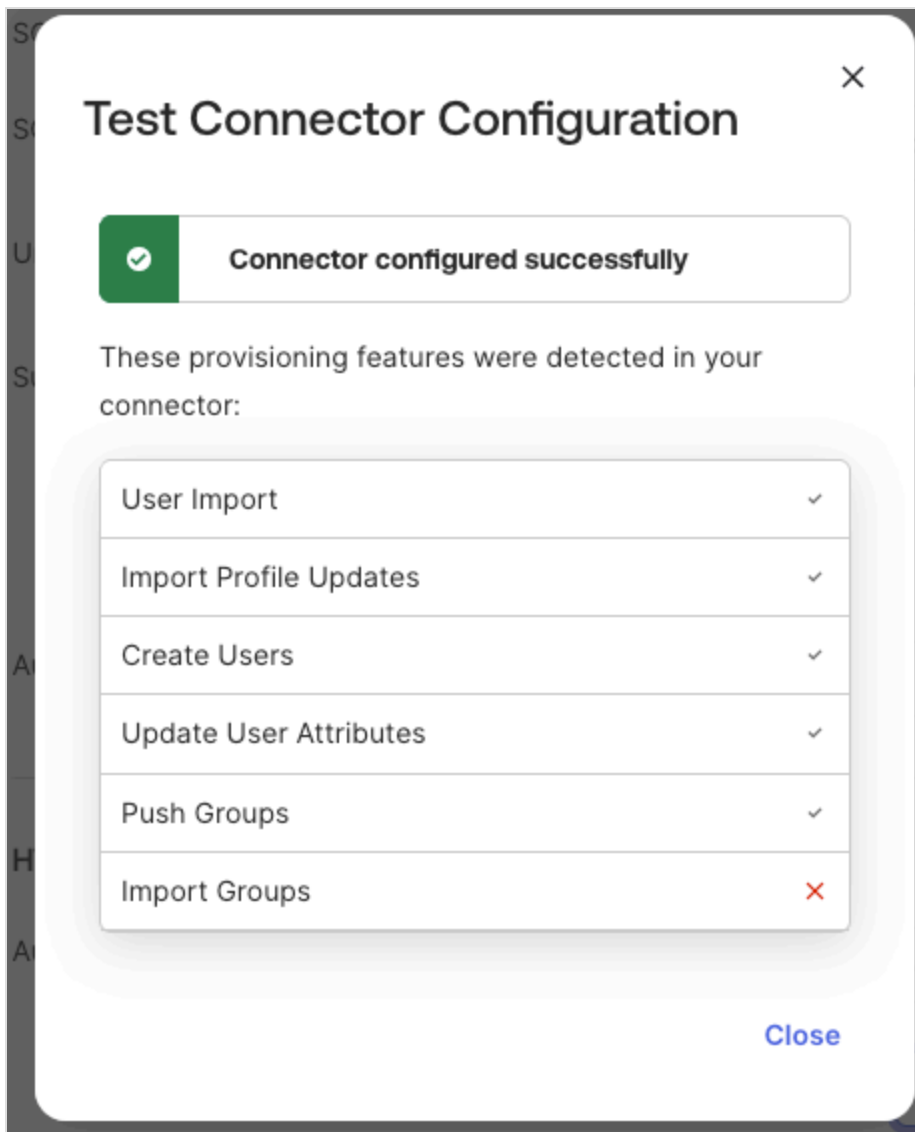
Staff, department group rules, the sign-on policy, the help desk role, and the MFA reset are Phase 1. [docs/phases/01-foundation.md](#). [docs/runbooks/mfa-reset.md](#).

Rostr is the app in `rostr/`. Staff sign in with SAML. Administrators sign in to `/admin/users` with OIDC. That page stays closed unless the token's groups include `APP-Rostr-Admins`. [docs/phases/02-onboarding.md](#).



Jonah Hale signed in as an administrator, with both Rostr groups in the token

Okta keeps the Rostr account in step with the directory over SCIM, at <https://rostr.lanternfieldgoods.co.uk/scim/v2>. When a mapped field changes, Okta sends the whole user back with PUT. Turning the account off is the same kind of PUT, with active set to false. Okta does not delete the Rostr user. [docs/phases/03-scim.md](#).



Okta's connector test against Rostr

scripts/hr-sync compares the HR file with the previous commit. It names the change as a joiner, a mover, or a leaver, and it does not call Okta unless `--apply` is passed. Three Okta workflows then do the directory write. Group rules put the person in the right department group and in APP-Rostr-Users. SCIM writes the same change into Rostr. In that test, Thomas Okeke was hired, Priya Shah moved to Operations, and Samir Adeyemi was deactivated. [docs/phases/04-jml.md](#). The script signs in as its own app, svc-jml-sync. The private key is not in the repo. [docs/decisions/scripts-identity.md](#).

Phase 5 is the governance work. Lena Ortiz asked for admin access in osTicket. A workflow added her to APP-Rostr-Admins and removed her an hour later. A separate review revoked two leftover memberships. A script listed accounts with no recent sign-in. Another script found an OAuth app that had been planted with broad access, and that access was revoked. The evidence pack is indexed to SOC 2 and ISO 27001 control names. This trial does not include Okta Identity Governance, so that index is a comparison, not a compliance claim. [docs/phases/05-governance.md](#). [docs/decisions/oig-mapping.md](#). [evidence/audit-pack/README.md](#).

Phase 6 broke six things on purpose, wrote down what the logs showed, and put each one back. [docs/incidents/](#).

Code in this repo was drafted with Cursor and reviewed before it was committed. Okta configuration, the tests, and these write-ups were done in the org.

Why

The design was written down before the clicking, because the user cap and the flow cap are real. Email is the username because Okta login, the SAML NameID, and the Rostr sign-in upsert are all the email. Okta's own guide advises against that. The lab kept it so the three identifiers stay the same, and recorded the trade-off. [docs/phases/o3-scim.md](#).

Apps are assigned to groups. Department groups are filled by rules. A membership added by hand does not follow a department change. Drill 6.5 is that fact, not a surprise. [docs/incidents/o6-mover-residue.md](#).

Stale-Access is a script rather than the fifth Workflow. The flow budget was already spent on Joiner, Mover, Leaver, and Access Request. A script can be re-run and diffed. A fifth flow could not be created without deleting one of those four.

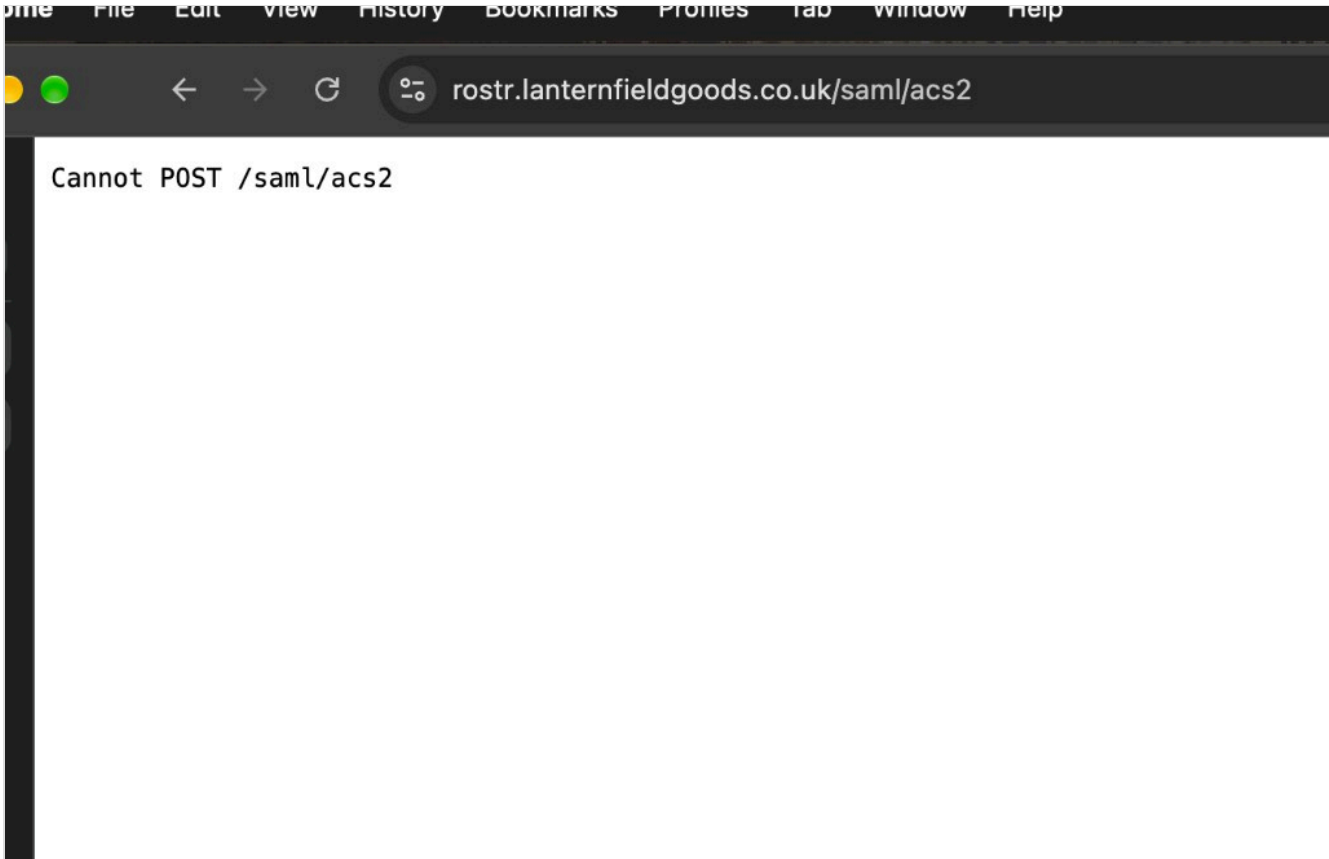
Risk management

Risk	What the lab actually did
Rostr is on the internet while the tunnel is up	SCIM requires a bearer token. A missing or wrong token is 401. The data is fictional. docs/phases/o3-scim.md
A secret in Git	.gitignore and .cursorignore block .env and keys. .env.example has names only. The service private key is under the home directory. docs/decisions/scripts-identity.md
Locked out of the org	Break-glass is a separate super admin. docs/decisions/o2-break-glass.md
Automation too powerful	svc-jml-sync has three scopes plus the app-grant scopes added for the OAuth review. Role and scope are separate. The review found a planted app and revoked its grants. docs/runbooks/oauth-review.md
A script changes the wrong person	hr-sync dry-runs unless --apply is passed, and it rejects a ticket that is not REQ- plus four digits.
The 10-user cap	Deactivated users do not count. Samir Adeyemi and Thomas Okeke are terminated in the HR file. Drill Scim was deactivated after 6.4.
The trial ends	This is a 30-day trial, not a 90-day idle Integrator org. Teardown has to decide what happens to the org before the trial converts. That decision is 7.4 and is not made here.
Over-claiming	This report does not say the lab administered a production org, owned a production lifecycle, or delivered a compliance programme.

Incidents

Record	What broke	What fixed it
oo	Two OIDC apps named Rostr Admin	Kept 00a18egjva6o5FpoP698. Deleted the duplicate.

01	osTicket staff login denied	The username was an md5 of an old email, and the password hash had to be bcrypt cost 8.
02	Single sign-on URL pointed at /saml/acs2	Browser stopped on Cannot POST /saml/acs2. The auth log gained no line. URL put back.



SAML posted to a route Rostr does not have

Record	What broke	What fixed it
03	Sign-in redirect URI did not match the client	Okta returned 400 invalid_request before Rostr saw the callback.
04	groups filter was APP-Rostrx	Sign-in succeeded and /me showed no groups. Filter put back.
05	Create required department, and the mapping was removed	Okta's retry returned 201 with no department. A later profile edit pushed Sales. Deactivation then set Rostr inactive.
06	Jonah was in APP-Rostr-Admins by hand, then moved to Operations	The department rules moved. The manual grant stayed until it was removed.
07	Deactivate Users was off	Thomas was DEPROVISIONED in Okta and still active in Rostr. The checkbox was turned back on, he was activated, and the second deactivate wrote active: false.


→


Edit

Provisioning to App

Create Users
☒ Enable

Creates or links a user in Rostr when assigning the app to a user in Okta.

The [default username](#) used to create accounts is set to **Okta username**.

Update User Attributes
☒ Enable

Okta updates a user's attributes in Rostr when the app is assigned. Future attribute changes made to the Okta user profile will automatically overwrite the corresponding attribute value in Rostr.

Deactivate Users
☐ Enable

Deactivates a user's Rostr account when it is unassigned in Okta or their Okta account is deactivated. Accounts can be reactivated if the app is reassigned to a user in Okta.

Sync Password
☐ Enable

Creates a Rostr password for each assigned user and pushes it to Rostr.

Deactivate Users off, which is why the first leaver never reached Rostr

The 6.1 success sign-in overwrote Jonah's planted stale `lastLogin`. The Phase 5 stale-access report no longer matches the database. [docs/incidents/o2-saml-acs.md](#).

Outcome

The lab is done when	State on 2026-10-07
One HR source drives joiner, mover, and leaver	Done. Thomas created, Priya moved, Samir deactivated. A second leaver on Samir changed nothing. docs/phases/o4-jml.md
SAML and OIDC can be explained from captures	Done. Jonah completed both. docs/phases/o2-onboarding.md
SCIM shows create, update, and deactivate	Done. The app sends PUT, including <code>active: false</code> . docs/phases/o3-scim.md
Group and attribute mappings are tested	Done, with a gap: a failed create's retry did not re-read a mapping that was fixed afterwards. The profile edit did. docs/incidents/o5-scim-mapping.md

An access request and a review	The grant and the hour-later remove are in the System Log. Ticket 357784 is Closed. REQ-0002 revoked leftover group membership. docs/phases/05-governance.md
An evidence pack	evidence/audit-pack/README.md
Six deliberate failures written up	Done. Incidents 02 through 07, plus the two earlier incidents.
Teardown	Not done.

Current HR file, which is ahead of the Phase 4 story: Jonah Hale is Operations, Priya Shah is Operations, Samir Adeyemi is terminated, Thomas Okeke is terminated. APP-Rostr-Admins is empty. An orphan row inserted straight into Rostr was imported, matched nobody, and was ignored. It is still in the database. <docs/phases/03-scim.md>.

What production would add

A real HRIS, not a JSON file. Okta Identity Governance for requests, certifications, and entitlements, instead of osTicket plus a Workflow plus a script. Device trust, which this lab did not build because it needs managed devices.

The Mover flow updates the profile and reads groups. It does not remove a membership that was added by hand. Production would have to decide whether that cleanup lives in the flow, in a review, or in a rule that forbids manual members of APP-* groups.

Deactivate Users has to stay on. Turning it off does not fail the Okta deactivation. It fails the app account, quietly, which is what 6.6 showed.

userName would not be the email if addresses change. This lab used the email on purpose.

The trial org, the ignored orphan, and the stale-access report that no longer matches lastLogin would not be left as the record of the directory. Ticket 357784 is closed. Teardown, or a renewal decision, is still outstanding.